

How Peakless reads ESB meter files

Test report, 9 October 2026, branch `test/meter-file-scenarios` off v8 at d32111d. Test and measurement only: the model and the way a file is reconciled with your answers are unchanged. Every number below comes from a run of the real app against homes whose true use and true costs are known.

Summary

- **The jump from about 1% to 65–87% came on 6 October, not from the tester fix.** Commit 123f0ba started reading meter files for when the panels went up. A home whose battery soaks up the winter's spare solar exports nothing until spring, so the months before were taken for a home without panels and the panels added on top: the tester-like home's price gap went from 2.8% to 86.6%. The tester fix (3a8346b) cut it to 59.6% but put a 4.5–14.3% gap on files with panels up all year and gave a no-battery home a 14.8-year payback (true 6.77). Today's build (e22e94a) is within 0.2–0.5% on all of them.
- **A full year from a home without panels reads well:** 11 of 12 full-year scenarios pass every tolerance; the one that does not is a heat pump home on calendar 2025 (-5.5% use), a different winter from the year ahead.
- **Five problems that make no sound.** A file that ends on the last day of a two-month period loses that period's neighbour (the last reading is stamped 00:00 the next day and counted as a whole day): nine months ending 28 February read -15.9% for the gas home. Rows that appear twice are counted twice (+99.5%); the same months five times, five times (+396.5%). US-style dates are half read with no warning. A typed yearly figure is replaced by the file without a word. The wrong panel count rebuilds the home's use from the wrong output (+49.9% use, payback 4.1 years against 6.77).
- **The accuracy figure does not move.** It reads $\pm 4\%$ to $\pm 7\%$ for every accepted file, including a four-week winter file that is +73.6% out and a doubled file that is +99.5% out. The import card does warn about short and partial files, once.
- **Short or seasonal files are fine for gas homes and far out for heat pumps:** four summer weeks -7.0% (gas) and -45.2% (heat pump); four winter weeks 0.0% and +73.6%.
- **Solar files without export rows (common in the first weeks) go wrong:** the app proposes a wrong install date, offers no answer that fits, reads use -34.2% and shows no payback.
- **Where the user's answers and the file disagree, the app mostly goes with one of them without saying so:** no solar but exports in the file (-34.2% use, no payback); EV or heat pump added partway (-17.2%, -17.4%); a file from a previous home (+87.1%).

- **A golden-households test now guards this.** 14 reference scenarios run through the real app on every push; on the 6 October build and the tester-fix build it fails the same five solar and battery homes, on today's it passes.

CI note: the brief says CI on v8 fails at the type check. That was fixed this morning (3a8346b); the last three pushes to v8 (3a8346b, e22e94a, d32111d) passed every step. CI runs on v8 and on pull requests, so the golden test starts guarding v8 when this branch is merged or opened as a pull request.

Step 0: the regression

The commits

Commit	When	What it did to a meter file from a home with panels
ad1de59	6 Oct 08:31	Panels up: the file is priced as recorded. The home's own use is taken to be what it bought, so the "without panels" side of every comparison, and the payback, is worked out on a home far smaller than it is.
123f0ba	6 Oct 09:25	Reads the file for when exports begin. If they begin more than two weeks in, the days before are taken to be the home without panels, the stated panels are added to them, and the days after are ignored unless they cover eleven months. A battery home exports nothing in winter, so a year with panels throughout became five winter months "without panels" with panels added again.
5179c9f	8 Oct	As 123f0ba for meter files.
3a8346b	9 Oct 09:01	The tester fix. With exports in the file, every day from the first export is "worked back" to the home's own use (bought + modelled output – sold – battery losses), and every plan, including the home's own system, is priced by simulating the panels again on that use. The home's use came right; its own bills became a model's estimate instead of the readings, and the misread start date stayed.
e22e94a	9 Oct 10:59	Panels up for the whole file and the stated system unchanged: priced from the readings again; the worked-back use is kept for the use figures and every what-if. If exports start partway, the app asks whether that is when the panels went up instead of assuming.

The benchmark homes from the five-way study, on each build

Same homes, same true costs, same frozen prices and date for every build. Price gap: the average gap between the app's yearly cost and the true cost across all 33 plans. € lost: what the app's top plan costs against the truly cheapest. Rows that changed between builds, plus two that did not.

Home (meter file)	ad1de59 6 Oct 08:31	123f0ba 6 Oct 09:25	5179c9f 8 Oct	3a8346b 9 Oct 09:01	e22e94a 9 Oct 10:59
Typical home, 4,200 kWh	0.12% · €0	0.12% · €0	0.12% · €0	0.12% · €0	0.12% · €0
Heat pump, 9,000 kWh	0.31% · €0	0.31% · €0	0.31% · €0	0.31% · €0	0.31% · €0
4 kWp solar, no battery	0.32% · €0	0.32% · €0	0.32% · €0	5.1% · €0	0.32% · €0

Home (meter file)	ad1de59 6 Oct 08:31	123f0ba 6 Oct 09:25	5179c9f 8 Oct	3a8346b 9 Oct 09:01	e22e94a 9 Oct 10:59
4 kWp solar + 10 kWh battery	0.39% · €0	81.31% · €0	83.62% · €101	61.23% · €101	0.39% · €0
6 kWp + 9 kWh battery, put in mid-April (file part before, part after)	120.79% · €58	26.76% · €202	21.93% · €101	24.09% · €101	24.09% · €101
6 kWp + 10 kWh battery + EV charged 2am-5am	17.36% · €32	75.63% · €32	73.44% · €32	36.34% · €32	0.38% · €0

The new ground-truth homes, on each build

Use: the app's yearly use against the true use. Gap: price gap as above. Payback: the app's against the true one ("none" where the app shows no payback).

Scenario	ad1de59	123f0ba	5179c9f	3a8346b	e22e94a
A1-gas: Calendar year 2025	-1.5% · 1.0% · 7.1/6.82	-1.5% · 1.0% · 7.1/6.82	-1.5% · 1.0% · 7.1/6.82	-1.5% · 1.0% · 7.1/6.82	-1.5% · 1.0% · 7.1/6.82
B2-gas_solar: A year after the panels, exports recorded	-34.2% · 0.4% · none/6.77	-34.2% · 0.4% · none/6.77	-34.2% · 0.3% · none/6.77	-1.8% · 4.5% · 14.8/6.77	-1.8% · 0.3% · 6.9/6.77
B2-hp_solar_batt: A year after the panels, exports recorded	-54.9% · 0.2% · none/6.2	-54.9% · 0.2% · none/6.2	-54.9% · 0.2% · none/6.2	-4.4% · 4.6% · 7.2/6.2	-4.4% · 0.2% · 7.2/6.2
B3-friend: Panels from 20 Oct 2025, exports recorded only from 1 Dec 2025 (the tester's case)	-40.0% · 2.8% · none/5.72	-19.4% · 86.6% · 5.4/5.72	-19.4% · 76.3% · 5.6/5.72	-15.0% · 59.6% · 8.7/5.72	+3.7% · 0.5% · 5.9/5.72
B4a-gas_solar: Before and after the panels (12 May 2025), exports recorded; user confirms the date	-20.1% · 56.5% · none/6.77	-1.8% · 2.0% · 7.0/6.77	-1.8% · 2.1% · 7.0/6.77	-2.9% · 3.0% · 9.9/6.77	-2.9% · 3.0% · 6.9/6.77
B4d-hp_solar_batt: Before and after a battery system (3 Mar 2025), exports recorded; user confirms the date shown	-37.6% · 49.3% · none/6.2	+3.8% · 10.7% · 5.5/6.2	+3.8% · 10.7% · 5.3/6.2	-5.3% · 3.5% · 6.4/6.2	-5.3% · 3.5% · 5.5/6.2
B5-hp_solar_gridfill: Battery filled from the grid at night in winter	-53.5% · 0.2% · none/5.97	-53.5% · 0.2% · none/5.97	-53.5% · 0.2% · none/5.97	-3.7% · 14.3% · 7.5/5.97	-3.7% · 0.2% · 7.7/5.97
B6-gas_solar: User says no solar; the file shows exports	-34.2% · 0.4% · none/6.77	-34.2% · 0.4% · none/6.77	-34.2% · 0.3% · none/6.77	-34.2% · 0.3% · none/6.77	-34.2% · 0.3% · none/6.77

Why: before 6 October the readings were priced as they are (good) but the home looked 34–55% smaller than it is, so no payback was shown, which is what the tester saw. 123f0ba broke the files where exports start late (the tester-like home, exports recorded from 1 December). 3a8346b fixed the size of the home but priced the home's own system by simulation, which costs 4–14% on files that were right before, and gave the no-battery home a 14.8-year payback. e22e94a keeps both right.

What "the app makes them match" does, and when

1. **Two-month totals (every file).** For each two-month period, the readings in it are averaged per day and multiplied by the period's length. Periods with no readings are filled from the average of the others, reshaped by the seasonal profile for the heating type. The import card shows the totals, marks filled periods with *, and says "Partial year" (under 300 days or a period missing) or "Only N days" (under 45 days).
2. **Where the panels are (files with exports or a midday drop, when the person says they have panels).** Exports from the first two weeks and the system unchanged: priced as recorded. Exports from later: asks "Did the panels go up around <date>?". No exports: asks "Is your meter file from before the panels went up?", or with a lasting midday drop, "Did the panels go up around <date>?".
3. **Worked-back use (files with panels in them).** The days after the panels are rebuilt as bought + the stated panels' modelled output – sold – 8% of the solar used, if there is a battery. The two-month totals are replaced with these, and periods the file does not cover are grown by the same ratio (used ÷ bought). This is what turns 2,600 kWh bought into a 4,000 kWh home. It runs on the stated system: a wrong panel count gives a wrong home.
4. **Only part of the file (a file that straddles the install, or a "no solar" answer with exports more than 60 days in).** Only the days before the panels are used, and the totals recomputed from them.
5. **Turning totals into hours.** With at least 330 days, each day of the year is the recorded day nearest the same date on the same weekday. With fewer, the two-month totals are spread on a 70/30 blend of the file's average day and the heating profile. Either way the year is rescaled to the totals.
6. **Typed figures.** Once a file is in, the typed yearly kWh and bill are ignored.

The test homes

Each home is two years and nine months of half-hour use (1 January 2024 to 8 October 2026), built from its own activity and Cork weather, not from the profiles Peakless uses. Solar output follows the day-to-day weather and is anchored month by month to PVGIS for Cork (south 977, south-west 919, north-east 621 kWh per kWp a year at 35°). The truth is the home as it is now over its last twelve months, priced as the year ahead on the plan list frozen on 9 October 2026, with announced price changes, standing charges, the PSO levy, export at each plan's rate and the tax on export income above €600. Solar saving and payback are counted the way the app counts them: the cheapest plan without the panels less the cheapest with them; the price after grant over that.

Home	Use, kWh	Bought / sold	Cheapest plan, true cost	System	Saving	Payback, years
Gas-heated home, 4,000 kWh	4,000	4,000 / 0	Energia Smart 24 Hour, €1,453	4.0 kWp, €6,000 after grant (quoted)	€880	6.82
Heat pump home, 7,500 kWh, heavy winter	7,500	7,500 / 0	Energia Smart 24 Hour, €2,476	4.0 kWp, €6,000 after grant (quoted)	€1,037	5.78
EV charged at night, 6,000 kWh	6,000	6,000 / 0	Flogas Smart EV Night Charge 29%, €1,785	4.0 kWp, €6,000 after grant (quoted)	€934	6.43
Gas home, 4 kWp south since 12 May 2025, no battery	4,000	2,639 / 2,585 (makes 3,946)	Electric Ireland Home Electric+ SST Saver 16%, €567	4.0 kWp, €6,000 after grant	€887	6.77
Heat pump home, 6 kWp + 10 kWh battery since 3 Mar 2025, battery on solar only	7,500	3,386 / 1,692 (makes 5,905)	Electric Ireland Home Electric+ SST Saver 16%, €846	6.0 kWp + 10 kWh, €10,100 after grant	€1,630	6.2
Heat pump home, 6 kWp + 10 kWh battery, filled from the grid at night Nov-Feb	7,500	3,490 / 1,745 (makes 5,905)	Electric Ireland Home Electric+ SST Saver 16%, €784	6.0 kWp + 10 kWh, €10,100 after grant	€1,692	5.97
The tester's case, rebuilt: heat pump 6,500 kWh, 22 panels on two faces, 9 kWh battery, panels from 20 Oct 2025, exports recorded from 1 Dec 2025	6,500	2,686 / 3,484 (makes 7,418)	Electric Ireland Home Electric+ SST Saver 16%, €332	9.7 kWp + 9 kWh, €10,600 after grant	€1,852	5.72
Gas home that bought an EV on 1 Apr 2026	6,400	6,400 / 0	Flogas Smart EV Night Charge 29%, €1,910	4.0 kWp, €6,000 after grant (quoted)	€939	6.39
Gas home that put in a heat pump on 15 Jan 2026	7,900	7,900 / 0	Energia Smart 24 Hour, €2,593	4.0 kWp, €6,000 after grant (quoted)	€1,036	5.79
Gas home, away 3-30 Aug 2026	4,000	4,000 / 0	Energia Smart 24 Hour, €1,453	4.0 kWp, €6,000 after grant (quoted)	€880	6.82

The tester's own file was not available. "The tester's case" is rebuilt from what he described: a heat pump home of about 6,500 kWh in the south, 12 panels south-west and 10 north-east, a 9 kWh battery, panels from 20 October 2025, and (as often happens) exports recorded by ESB only from 1 December. With his file, MPRN and serial removed, it can be added as a scenario.

Results: every scenario

Tolerances (the brief's): use and bill within 5%, payback within half a year, the best plan or one within €25 a year. For partial and short files the seasons have to be guessed, and we accept 10% (nine months, this year so far) and 15% (two to six weeks) on use and bill, 0.75 and 1 year on payback: wider than that and a plan or

payback decision could flip. Bill: the app's yearly cost for its top plan against that plan's true cost. For a home planning panels, plans are compared as the home is today. For a battery that only stores solar, the app's answer for that (given under its headline) is scored; its headline assumes night filling. Accuracy: the "±" figure the app shows.

ID	Scenario	What the app did	Use	Top plan (€ over cheapest)	Bill	Saving	Payback app / true	Shown	Result
A1-gas	Calendar year 2025	accepted	-1.5%	Energia Smart 24 Hour	-0.9%	-3.9%	7.1 / 6.82	±4%	pass
A1-heatpump	Calendar year 2025	accepted	-5.5%	Energia Smart 24 Hour	-4.6%	-5.1%	6.1 / 5.78	±4%	FAIL: consumption
A1-ev	Calendar year 2025	accepted	-0.8%	Flogas Smart EV Night Charge 29%	-0.1%	-4.0%	6.7 / 6.43	±4%	pass
A2-gas	Twelve months, March 2025 to February 2026	accepted	-0.9%	Energia Smart 24 Hour	-0.5%	-3.9%	7.1 / 6.82	±4%	pass
A2-heatpump	Twelve months, March 2025 to February 2026	accepted	+0.5%	Energia Smart 24 Hour	+0.6%	-4.7%	6.1 / 5.78	±4%	pass
A2-ev	Twelve months, March 2025 to February 2026	accepted	+1.8%	Flogas Smart EV Night Charge 29%	+0.7%	-4.2%	6.7 / 6.43	±4%	pass
A3a-gas	Nine months, June 2025 to February 2026 (no spring)	warned (partial year)	-15.9%	Energia Smart 24 Hour	-12.8%	-3.5%	7.1 / 6.82	±4%	FAIL: consumption, bill
A3a-heatpump	Nine months, June 2025 to February 2026 (no spring)	warned (partial year)	-17.8%	Energia Smart 24 Hour	-15.8%	-7.9%	6.3 / 5.78	±4%	FAIL: consumption, bill
A3a-ev	Nine months, June 2025 to February 2026 (no spring)	warned (partial year)	-13.3%	Electric Ireland Home Electric+ SST Saver 16% (+€50)	-11.5%	-3.9%	6.7 / 6.43	±4%	FAIL: consumption, bill, plan
A3b-gas	Nine months, September 2025 to May 2026 (no summer)	warned (partial year)	-0.9%	Energia Smart 24 Hour	-0.7%	-1.5%	6.9 / 6.82	±4%	pass
A3b-heatpump	Nine months, September 2025 to May 2026 (no summer)	warned (partial year)	+8.1%	Energia Smart 24 Hour	+7.2%	+0.3%	5.8 / 5.78	±4%	pass
A3b-ev	Nine months, September 2025 to May 2026 (no summer)	warned (partial year)	-2.6%	Electric Ireland Home Electric+ SST Saver 16% (+€50)	-1.8%	-1.8%	6.5 / 6.43	±4%	FAIL: plan
A4-gas	This year so far, 1 January to 8 October 2026	warned (partial year)	0.0%	Energia Smart 24 Hour	0.0%	-1.3%	6.9 / 6.82	±4%	pass
A4-heatpump	This year so far, 1 January to 8 October 2026	warned (partial year)	-3.3%	Energia Smart 24 Hour	-2.9%	-3.3%	6.0 / 5.78	±4%	pass
A4-ev	This year so far, 1 January to 8 October 2026	warned (partial year)	+0.6%	Electric Ireland Home Electric+ SST Saver 16% (+€50)	+0.8%	-1.1%	6.5 / 6.43	±4%	FAIL: plan
A5a-gas	Four summer weeks, July 2026	warned (short file)	-7.0%	Energia Smart 24 Hour	-5.6%	-3.7%	7.1 / 6.82	±4%	pass

ID	Scenario	What the app did	Use	Top plan (€ over cheapest)	Bill	Saving	Payback app / true	Shown	Result
A5a-heatpump	Four summer weeks, July 2026	warned (short file)	-45.2%	Energia Smart 24 Hour	-40.0%	-13.2%	6.7 / 5.78	±4%	FAIL: consumption, bill
A5a-ev	Four summer weeks, July 2026	warned (short file)	-9.1%	Electric Ireland Home Electric+ SST Saver 16% (+€50)	-7.3%	-1.5%	6.5 / 6.43	±4%	FAIL: plan
A5b-gas	Four winter weeks, January 2026	warned (short file)	0.0%	Energia Smart 24 Hour	0.0%	-2.6%	7.0 / 6.82	±4%	pass
A5b-heatpump	Four winter weeks, January 2026	warned (short file)	+73.6%	Energia Smart Data (+€15)	+63.6%	+10.9%	5.2 / 5.78	±4%	FAIL: consumption, bill
A5b-ev	Four winter weeks, January 2026	warned (short file)	-2.1%	Electric Ireland Home Electric+ SST Saver 16% (+€50)	-1.4%	-2.3%	6.6 / 6.43	±4%	FAIL: plan
A5c-gas	Two summer weeks, August 2026	warned (short file)	-8.4%	Energia Smart 24 Hour	-6.8%	-0.4%	6.8 / 6.82	±4%	pass
A5c-heatpump	Two summer weeks, August 2026	warned (short file)	-41.5%	Energia Smart 24 Hour	-36.8%	-11.7%	6.6 / 5.78	±4%	FAIL: consumption, bill
A5c-ev	Two summer weeks, August 2026	warned (short file)	-12.2%	Electric Ireland Home Electric+ SST Saver 16% (+€50)	-9.1%	-0.9%	6.5 / 6.43	±4%	FAIL: plan
A6-gas	An old year, July 2024 to June 2025	accepted	-1.6%	Energia Smart 24 Hour	-1.2%	-4.1%	7.1 / 6.82	±4%	pass
A6-heatpump	An old year, July 2024 to June 2025	accepted	-4.2%	Energia Smart 24 Hour	-3.7%	-4.5%	6.1 / 5.78	±4%	pass
A6-ev	An old year, July 2024 to June 2025	accepted	-0.1%	Flogas Smart EV Night Charge 29%	-0.3%	-4.9%	6.8 / 6.43	±4%	pass
A7-gas	Two years, October 2024 to October 2026	accepted	-0.8%	Energia Smart 24 Hour	0.0%	-3.7%	7.1 / 6.82	±4%	pass
A7-heatpump	Two years, October 2024 to October 2026	accepted	-2.0%	Energia Smart 24 Hour	+0.1%	-5.0%	6.1 / 5.78	±4%	pass
A7-ev	Two years, October 2024 to October 2026	accepted	+0.6%	Flogas Smart EV Night Charge 29%	0.0%	-4.5%	6.7 / 6.43	±4%	pass
B1a-friend	Full year before the panels	asked "Is your meter file from before the panels went up?"	-4.5%	Electric Ireland Home Electric+ SST Saver 16%	-46.3%	+12.2%	5.1 / 5.72	±4%	FAIL: bill, payback
B1b-friend	Half year before the panels	warned (partial year); asked "Is your meter file from before the panels went up?"	-23.7%	Electric Ireland Home Electric+ SST Saver 16%	-145.9%	+6.9%	5.4 / 5.72	±4%	FAIL: consumption, bill
B1c-friend	Summer before the panels	warned (partial year); asked "Is your meter file from before the panels went up?"	-55.6%	Pinergy Lifestyle Family Time (+€245)	-220.1%	-0.8%	5.8 / 5.72	±4%	FAIL: consumption, bill, plan

ID	Scenario	What the app did	Use	Top plan (€ over cheapest)	Bill	Saving	Payback app / true	Shown	Result
B1d-friend	Winter before the panels	warned (partial year); asked "Is your meter file from before the panels went up?"	-14.6%	Electric Ireland Home Electric+ SST Saver 16%	-77.7%	+5.5%	5.4 / 5.72	±4%	FAIL: consumption, bill
B2-gas_solar	A year after the panels, exports recorded	accepted	-1.8%	Electric Ireland Home Electric+ SST Saver 16%	0.0%	-2.3%	6.9 / 6.77	±4%	pass
B3-gas_solar	A year after the panels, no export rows	asked "Did the panels go up around 22 Jan 2026?"	-34.2%	Energia Smart 24 Hour (+€11)	+83.0%	—	none / 6.77	±4%	FAIL: consumption, bill, no payback shown
B2-hp_solar_batt	A year after the panels, exports recorded	accepted	-4.4%	Electric Ireland Home Electric+ SST Saver 16%	+0.1%	-14.3%	7.2 / 6.2	±4%	FAIL: payback
B3-hp_solar_batt	A year after the panels, no export rows	asked "Did the panels go up around 5 Feb 2026?"	-54.9%	Electric Ireland Home Electric+ SST Saver 16%	+39.1%	—	none / 6.2	±4%	FAIL: consumption, bill, no payback shown
B3-friend	Panels from 20 Oct 2025, exports recorded only from 1 Dec 2025 (the tester's case)	asked "Did the panels go up around 27 Feb 2026?"	+3.7%	Electric Ireland Home Electric+ SST Saver 16%	+0.5%	-3.0%	5.9 / 5.72	±4%	pass
B4a-gas_solar	Before and after the panels (12 May 2025), exports recorded; user confirms the date	asked "Did the panels go up around 12 May 2025?"	-2.9%	Electric Ireland Home Electric+ SST Saver 16%	-1.9%	-2.5%	6.9 / 6.77	±5%	pass
B4b-gas_solar	Same file; user says the panels were up the whole time	asked "Did the panels go up around 12 May 2025?"	+42.2%	Energia Smart 24 Hour (+€11)	+67.1%	+10.6%	6.1 / 6.77	±4%	FAIL: consumption, bill, payback
B4c-gas_solar	Before and after the panels, no export rows; user confirms the date shown	asked "Did the panels go up around 14 May 2025?"	-3.0%	Electric Ireland Home Electric+ SST Saver 16%	-0.7%	-3.5%	7.0 / 6.77	±4%	pass
B4d-hp_solar_batt	Before and after a battery system (3 Mar 2025), exports recorded; user confirms the date shown	asked "Did the panels go up around 5 Mar 2025?"	-5.3%	Electric Ireland Home Electric+ SST Saver 16%	-6.9%	+12.3%	5.5 / 6.2	±5%	FAIL: consumption, bill, payback
B5-hp_solar_gridfill	Battery filled from the grid at night in winter	accepted	-3.7%	Electric Ireland Home Electric+ SST Saver 16%	+0.2%	-22.4%	7.7 / 5.97	±4%	FAIL: payback
B6-gas_solar	User says no solar; the file shows exports	accepted	-34.2%	Electric Ireland Home Electric+ SST Saver 16%	0.0%	—	none / 6.77	±7%	FAIL: consumption, no payback shown
B7a-gas	User says they have 9 panels; the file shows a home without them (they have none)	asked "Is your meter file from before the panels went up?"	-0.2%	Energia Smart 24 Hour	0.0%	—	none / 6.82	±4%	FAIL: no payback shown

ID	Scenario	What the app did	Use	Top plan (€ over cheapest)	Bill	Saving	Payback app / true	Shown	Result
B7b-gas	Same; user answers that the file is from before the panels	asked "Is your meter file from before the panels went up?"	-0.2%	Energia Smart 24 Hour	-58.3%	-3.7%	7.1 / 6.82	±4%	FAIL: bill
B8-gas_solar	User says 14 panels; the home has 9	accepted	+49.9%	Electric Ireland Home Electric+ SST Saver 16%	0.0%	+65.9%	4.1 / 6.77	±4%	FAIL: consumption, payback
C1-gas_ev_now	EV bought on 1 Apr 2026, halfway through the file	accepted	-17.2%	Electric Ireland Home Electric+ SST Saver 16% (+€51)	-10.1%	-2.3%	6.5 / 6.39	±4%	FAIL: consumption, bill, plan
C2-gas_to_hp	Heat pump put in on 15 Jan 2026 (was gas)	accepted	-17.4%	Energia Smart 24 Hour	-15.4%	-8.7%	6.3 / 5.79	±4%	FAIL: consumption, bill
C4-gas_holiday	Away for four weeks in August 2026	accepted	-5.2%	Energia Smart 24 Hour	-4.0%	-4.6%	7.2 / 6.82	±4%	FAIL: consumption
C3-moved	Moved house: the file is the old (heat pump) home; the new home is gas-heated	accepted	+87.1%	Energia Smart 24 Hour	+70.5%	+12.0%	6.1 / 6.82	±4%	FAIL: consumption, bill, payback
D1a-gas	Daily register file instead of half-hourly	Rejected: This is one of ESB's daily files: a reading a day, without the hours that decide most plans	—	—	—	—	—	—	n/a
D1b-gas	Half-hourly file in kWh, not kW	accepted	-0.3%	Energia Smart 24 Hour	0.0%	-3.7%	7.1 / 6.82	±4%	pass
D1c-gas	Day and night register file (24-hour or night-storage meter)	Rejected: This is one of ESB's daily files: a reading a day, without the hours that decide most plans	—	—	—	—	—	—	n/a
D2a-gas	Edited in Excel: US dates (MM/DD)	warned (partial year)	-1.5%	Energia Smart 24 Hour	-1.2%	-1.0%	6.9 / 6.82	±4%	pass
D2b-gas	Edited in Excel: semicolons and comma decimals	Rejected: Only 0 readings found. Is this the file from ESB Networks?	—	—	—	—	—	—	n/a
D2c-gas	Header row renamed	Rejected: Only 0 readings found. Is this the file from ESB Networks?	—	—	—	—	—	—	n/a
D2d-gas	Saved by Excel with Irish short dates (d/m/yyyy h:mm)	accepted	-0.2%	Energia Smart 24 Hour	0.0%	-3.7%	7.1 / 6.82	±4%	pass
D3a-gas	Missing days: ten single days and a three-week gap	accepted	-0.8%	Energia Smart 24 Hour	-2.6%	-4.2%	7.1 / 6.82	±4%	pass
D3b-gas	Every row twice	accepted	+99.5%	Energia Smart 24 Hour	+80.3%	+4.1%	6.6 / 6.82	±4%	FAIL: consumption, bill
D3c-gas	Two overlapping files uploaded one after the other	warned (partial year)	-0.8%	Energia Smart 24 Hour	0.0%	-3.7%	7.1 / 6.82	±4%	pass
D3d-gas	Meter replaced in March: two serial numbers in one file	accepted	-0.2%	Energia Smart 24 Hour	0.0%	-3.7%	7.1 / 6.82	±4%	pass
D4-gas	Calendar 2024: 29 February and both clock changes	accepted	-1.2%	Energia Smart 24 Hour	-1.0%	-4.0%	7.1 / 6.82	±4%	pass
D5a-gas	Empty file	Rejected: File appears empty or invalid	—	—	—	—	—	—	n/a

ID	Scenario	What the app did	Use	Top plan (€ over cheapest)	Bill	Saving	Payback app / true	Shown	Result
D5b-gas	Header only	Rejected: File appears empty or invalid	—	—	—	—	—	—	n/a
D5c-gas	A bank statement CSV	Rejected: This is one of ESB's daily files: a reading a day, without the hours that decide most plans	—	—	—	—	—	—	n/a
D5d-gas	A gas meter file	Rejected: This is one of ESB's daily files: a reading a day, without the hours that decide most plans	—	—	—	—	—	—	n/a
D5e-gas	A PDF bill renamed .csv	Rejected: Only 0 readings found. Is this the file from ESB Networks?	—	—	—	—	—	—	n/a
D5f-gas	A very large file (the same 33 months five times, about 14 MB)	accepted	+396.5%	Energia Smart 24 Hour	+321.4%	+15.0%	5.9 / 6.82	±4%	FAIL: consumption, bill, payback
D6-gas	Typed 5,600 kWh in setup, then uploaded a file showing 4,000	accepted	-0.2%	Energia Smart 24 Hour	0.0%	-3.7%	7.1 / 6.82	±4%	pass

By group: A: 18 pass, 12 fail, 0 rejected; B: 4 pass, 14 fail, 0 rejected; C: 0 pass, 4 fail, 0 rejected; D: 8 pass, 2 fail, 9 rejected.

What the user sees

✓ Imported 13,106 readings

Jan-Feb: 697 kWh · Mar-Apr: 4 kWh · May-Jun: 617 kWh ·
Jul-Aug: 634 kWh · Sep-Oct: 692 kWh · Nov-Dec: 721 kWh
Total: 3,365 kWh/yr — anticipated full-year profile
Readings in kW (avg per 30-min interval) — converted ×0.5 to kWh

Partial year: 274 days across 6 of 6 billing periods. We filled the gaps from the rest. A full year would make the seasons more accurate.

Your file runs 1 Jun 2025 to 28 Feb 2026. No solar shows in it.

Use this data →

Nine months ending 28 February: March–April shows 4 kWh and "6 of 6 billing periods"; the year reads -15.9%.

✓ Imported 1,344 readings

Jan-Feb: 2,191 kWh · Mar-Apr: 2,280 kWh* · May-Jun: 2,007 kWh* · Jul-Aug: 1,958 kWh* · Sep-Oct: 2,075 kWh* ·
Nov-Dec: 2,510 kWh*
Total: 13,021 kWh/yr — anticipated full-year profile
Readings in kW (avg per 30-min interval) — converted ×0.5 to kWh
* estimated, no readings for this period

⚠ Only 29 days of data . That's well short of a year, so the seasons are a guess. Treat these figures as rough, and add a full year when you can.

Your file runs 5 Jan 2026 to 1 Feb 2026. No solar shows in it.

Use this data →

Four winter weeks of a heat pump home: warned once, then ±4% on every figure; the year reads +73.6%.

Step 14 of 20 · Your usage

Did the panels go up around 22 Jan 2026?

Your meter file shows daytime buying dropping from then, with nothing sent to the grid.

Yes, around then
We'll use the readings after that date

No, it's from before them
We'll add the panels to it

Panels up all year, ESB not recording exports: a wrong date, and neither answer is true.

Step 14 of 20 · Your usage

Did the panels go up around 12 May 2025?

Your meter file shows power sent to the grid from then. A battery can hold winter's spare solar, so the panels may have been up before.

Yes, around then

The days before are your home without panels



No, they were up for the whole file

We'll use your readings as they are



A file that straddles the install: the right date proposed, and the right answer available.

✗ Parse error

This is one of ESB's daily files: a reading a day, without the hours that decide most plans. In your ESB Networks account, open My Meter, then Downloads, and choose "30-minute readings in kW" (the kWh version works too).

A bank statement is called "one of ESB's daily files".

✓ Imported 35,040 readings

Jan-Feb: **1,394 kWh** · Mar-Apr: **1,329 kWh** · May-Jun: **1,176 kWh** · Jul-Aug: **1,275 kWh** · Sep-Oct: **1,363 kWh** · Nov-Dec: **1,441 kWh**

Total: 7,978 kWh/yr — anticipated full-year profile

Readings in kW (avg per 30-min interval) — converted ×0.5 to kWh

✓ A full year of readings.

Your file runs 9 Oct 2025 to 8 Oct 2026. No solar shows in it.

Use this data →

Every row twice: "A full year of readings", +99.5%, no warning.

✓ Imported 6,912 readings

Jan-Feb: **625 kWh** · Mar-Apr: **703 kWh** · May-Jun: **630 kWh** · Jul-Aug: **678 kWh** · Sep-Oct: **646 kWh** · Nov-Dec: **657 kWh**

Total: 3,939 kWh/yr — anticipated full-year profile

Readings in kW (avg per 30-min interval) — converted ×0.5 to kWh

Partial year: 145 days across 6 of 6 billing periods. We filled the gaps from the rest. A full year would make the seasons more accurate.

Use this data →

US-style dates: 6,912 of 17,520 readings read, shown as a "Partial year".

✓ Imported 17,520 readings

Jan-Feb: **697 kWh** · Mar-Apr: **664 kWh** · May-Jun: **588 kWh** · Jul-Aug: **638 kWh** · Sep-Oct: **682 kWh** · Nov-Dec: **721 kWh**

Total: 3,990 kWh/yr — anticipated full-year profile

Readings in kW (avg per 30-min interval) — converted ×0.5 to kWh

✓ A full year of readings.

Your file runs 9 Oct 2025 to 8 Oct 2026. No solar shows in it.

Use this data →

5,600 kWh typed in setup, then this file: the typed figure is replaced without a word.

User errors: how the app handles them

What the person did	What the app did	What it says about trust	How wrong
Says the panels were up all along; they went up in May	asked "Did the panels go up around 12 May 2025?"	±4%	use +42.2%, bill +67.1%, payback 6.1 vs 6.77
Says no solar; the file shows exports from the first day	accepted	±7% (says "Meter file shows panels already running")	use -34.2%, bill 0.0%, payback none vs 6.77
Says 9 panels; there are none	asked "Is your meter file from before the panels went up?"	±4%	use -0.2%, bill 0.0%, payback none vs 6.82
Says 9 panels and that the file is from before them; there are none	asked "Is your meter file from before the panels went up?"	±4%	use -0.2%, bill -58.3%, payback 7.1 vs 6.82
Says 14 panels; there are 9	accepted	±4%	use +49.9%, bill 0.0%, payback 4.1 vs 6.77
Uploads the file from the house they left	accepted	±4%	use +87.1%, bill +70.5%, payback 6.1 vs 6.82
Typed 5,600 kWh, then uploads a file showing 4,000	accepted	±4%	use -0.2%, bill 0.0%, payback 7.1 vs 6.82
Opened and saved the file in Excel with US dates	warned (partial year)	±4%	use -1.5%, bill -1.2%, payback 6.9 vs 6.82
File with every row twice	accepted	±4%	use +99.5%, bill +80.3%, payback 6.6 vs 6.82
Two overlapping files, one after the other	warned (partial year)	±4%	use -0.8%, bill 0.0%, payback 7.1 vs 6.82
Excel with semicolons and comma decimals	Rejected: Only 0 readings found. Is this the file from ESB Networks?	Turned away	—
Header row renamed	Rejected: Only 0 readings found. Is this the file from ESB Networks?	Turned away	—
A bank statement	Rejected: This is one of ESB's daily files: a reading a day, without the hours that decide most plans	Turned away	—
A gas meter file	Rejected: This is one of ESB's daily files: a reading a day, without the hours that decide most plans	Turned away	—
A PDF renamed .csv	Rejected: Only 0 readings found. Is this the file from ESB Networks?	Turned away	—
An empty file	Rejected: File appears empty or invalid	Turned away	—

Questions answered

1. How each group is handled today

- **No panels yet, last year's file (A).** The file is read into six two-month totals (per-day average $\times$ days in the period), missing periods filled from the others on the seasonal profile; with 330+ days the year is the real days, otherwise the totals spread on an average day. Full years land within 0.1–5.5% on use. Partial years are right when they end mid-period and wrong when they end on a period's last day (the midnight bug). Under 330 days the EV home loses its night charging pattern and is offered the wrong plan (+€50 a year). Two years: the per-day averages use both, the hours use the latest.
- **Panels already up (B).** If the file shows exports from the start and the stated system has not changed, the readings are priced as they are and the home's use is worked back. If exports start later, or there are none, it asks one question, the answer decides which days are 'before' and which are priced. With a battery, setup does not ask how it charges once a file is in, and the headline assumes it is filled at night.
- **The home changed in the file (C).** Nothing looks for it. An EV or heat pump added partway is averaged into the year (about 17% low), a holiday is averaged in, and a file from a previous home is taken as this one.
- **File problems (D).** Daily files, empty files and files it cannot parse are turned away with a message (two of the messages are wrong about what the file is). kWh files, gaps, a replaced meter, two overlapping uploads, the leap day and both clock changes are handled. Duplicated rows and US dates get through silently.

2. What period the figures represent

A modelled year: the file's days (or its two-month totals) laid on a calendar year, priced at today's prices with announced changes from their dates. In effect, "the next twelve months, if they are like the file's year". Home and Plans say "a year"; a few places say "a typical year"; the import card says "anticipated full-year profile" and, for files over 400 days, "averaged per day, so the result is one typical year". Nothing says which months the figures stand for or that weather is not adjusted. A person should expect the twelve-month total of their bills to match within the error shown here when the coming year is like the file's; individual two-month bills will not match, because winter and summer differ.

3. How a partial or seasonal file becomes a year

Each two-month period with readings is its per-day average $\times$ its length. A period with none is the average of the covered periods $\times$ (that period's share in the seasonal profile $\div$ the average share). Gas homes use ESB's standard load profile (urban or rural); heat pump and storage homes their own winter-heavy profiles. For a gas home this is close: four weeks of summer -7.0%, four weeks of winter 0.0%. For a heat pump it is far out:

summer -45.2%, winter +73.6%, because the profile's winter-to-summer ratio is much flatter than a real heat pump's and nothing adjusts for the weather of the weeks in the file.

4. Solar homes: rebuilding use from imports, and missing exports

Use is worked back month by month: what was bought, plus what the stated panels make in a typical year for that month, less what was sold, less 8% of the solar used where there is a battery. Without a battery the night hours are the meter's own and the solar used is put into daylight hours; with one, each day's total is spread on the home's own shape from before the panels (if there are three weeks of it) blended with the heating profile. It is right when the stated system is right (-1.8% for the 4 kWp home) and wrong in step with the system when it is not (+49.9% with 14 panels stated for 9). With no export rows, the app either asks whether the file is from before the panels or, if the midday buying drops, proposes a date; answering that the system never exports prices the readings as they are, reads use as what was bought (-34.2% and -54.9%), leaves out the export income the coming year will pay (bill +83.0%) and shows no payback.

5. Where the app changes the data or the answers without showing it

- Fills missing two-month periods from the others (shown with * on the import card, not afterwards).
- Counts a lone midnight reading as a whole day of the next period (the bug above): silent.
- Adds up rows that appear twice: silent.
- Reads US-style dates as Irish ones, dropping the days that cannot be months: shown only as "Partial year".
- Merges a new upload with earlier ones (latest 400 days kept) while the yearly totals come from the newest file alone: silent.
- Replaces a typed yearly kWh or bill with the file: silent.
- Rebuilds the home's use from the stated panels, and grows the periods the file does not cover by the same ratio: one line in the accuracy panel ("worked back").
- Uses only the days before the panels when a file straddles them, or when the person says no solar but exports appear after 60 days: the import card says which days; the accuracy panel says "Meter file shows panels already running".
- Assumes a battery is filled from the grid at night when a file is in (the question is skipped): the Plans page says so and gives the solar-only answer.
- Credits export income from the file to a home said to have no panels: silent.
- Skips readings below 0 or above 50 kW: silent (the count is not shown).

- Moves each day to the nearest same weekday within three days and takes the latest year when there are two: silent, and harmless in these runs.

Risk register

Risk	When it happens	How wrong	How likely	Today	Proposed fix
Last reading of a period counted as a day of the next	A file ending on the last day of Feb, Apr, Jun, Aug, Oct or Dec (a calendar-year download ends 31 Dec)	-15.9% to -17.8% (nine months); -1.5% (calendar year)	High	Silent	Date each reading by the start of its half hour in the totals, as the hourly ledger already does
Duplicate rows counted twice	Files joined in Excel, re-downloads appended	+99.5%; +396.5% for five copies	Low–medium	Silent, "A full year of readings"	Keep one reading per half hour and type; say how many duplicates were dropped
Short or seasonal file for a heat pump home	New meters, recent movers, partial downloads	-45.2% to +73.6%	Medium	Warned once; accuracy stays $\pm 4\%$	Scale with degree days for heat pumps; widen the accuracy figure with coverage; ask for a bill to anchor
Accuracy figure ignores coverage and contradictions	Any short, partial, doubled or contradicted file	Shows $\pm 4\%$ on results up to +396.5% out	High	—	Work the file part of the accuracy figure out from days covered, periods filled, duplicates and unresolved questions
Under 330 days loses when things happen	EV, night storage, any timed load with a partial file	Wrong plan, +€50 a year (EV)	Medium	Silent	Spread each month on that month's own recorded days instead of one blended average day
Solar file without export rows	First weeks after an install; export not yet registered	use -34.2%, bill +83.0%, no payback	High for new installs	Wrong date proposed; no answer fits	Add "Panels were up the whole time; exports aren't on the file yet"; model the exports for the coming year
No solar said, exports in the file	Person unsure, or forgot	use -34.2%, no payback	Medium	Accuracy panel only	Ask: "Your file shows power sent to the grid from <date>. Do you have panels?"
Wrong panel count	Typing from memory	use +49.9%, payback 4.1 vs 6.77	Medium	Silent	Compare expected and recorded exports; ask when they differ by more than about a third
Home changed in the file	EV bought, heat pump fitted, extension	-17.2% / -17.4%; EV plan +€51	Medium	Silent	Look for a lasting step in use or in night use; ask "Did something change around <date>?" and price the part after
File from another home	Moved house, landlord's file	+87.1%	Low	Silent	One question after upload: "Is this file from the home you live in now?"
Typed figure replaced by the file	Typed first, uploaded later	The figure the person gave is dropped	Medium	Silent	Say "Your file says 3,990 kWh; you told us 5,600. We'll use the file." with an undo

Risk	When it happens	How wrong	How likely	Today	Proposed fix
Misread or misnamed files	Excel edits, wrong downloads	US dates half read; semicolon and renamed-header files refused as "0 readings"; a bank or gas file called an ESB daily file	Medium	Mixed	Accept semicolons and comma decimals; spot day-month order; find columns by content; name what the file looks like
Battery charging assumed with a file	Battery homes uploading a file	headline plan +€118 a year if the battery is not filled at night (tester's case)	Medium	Plans page gives both answers; setup does not ask	Ask how the battery charges whatever the input
Weather of the file's year	Heat pump homes, any single year	-5.5% (calendar 2025)	High, small	Not adjusted, not said	Degree-day adjustment for heating; say which months the figures stand for

Proposed fixes, ranked

Impact is how much wrong answers would fall, across how many people; effort is a rough size. None is implemented; each needs your approval. Each should come with its scenario added to the golden households, and the golden limits tightened where it closes a gap.

#	Fix	Impact	Effort	Scenarios it fixes
1	Date readings by the start of their half hour in the two-month totals	High	Small	A3a, A1 (and every file ending on a period boundary)
2	Drop duplicate readings and say so	High when it happens	Small	D3b, D5f
3	Make the accuracy figure reflect coverage, filled periods, duplicates and open questions	High	Small	A3–A5, B3, B6, C, D
4	Ask instead of overriding: typed vs file, no-solar-but-exports, "is this file from this home?"	Medium–high	Small	D6, B6, C3
5	Solar file without exports: a true answer option, and expected exports modelled	High for new installs	Medium	B3
6	Ask how a battery charges also when a file is uploaded	Medium	Small	B1, B4d
7	Cross-check the stated panels against recorded exports	Medium	Medium	B8, B7
8	Partial files: month-specific days for timing; degree days for heat pumps	Medium–high	Medium–large	A3–A5 (EV and heat pump), C4
9	Spot a step change in the file and ask	Medium	Medium	C1, C2
10	Parser: semicolons, comma decimals, day-month order, columns by content, clearer refusals	Low–medium	Small	D2a–D2c, D5c, D5d
11	Weather adjustment and saying which months the figures stand for	Low–medium	Large	A1-heatpump

The golden households

`tests/e2e/golden-households.spec.js` runs 14 scenarios through the real app with the plan list and the date frozen, and compares with the stored truth. Limits are in `tests/fixtures/meter-scenarios/golden.json`: the agreed tolerance where the app meets it today; where it does not yet, today's error with a 5% margin, marked as a known gap, so a change can only make it better. Run on the 6 October build and on the tester-fix build, it fails B2-gas_solar, B2-hp_solar_batt, B3-friend, B4a-gas_solar and B5-hp_solar_gridfill; on today's build all 14 pass.

Scenario	Limits (use / bill / plan / payback)	Known gap
A1-gas	5.0% / 5.0% / €25 / 0.5 y	
A2-heatpump	5.0% / 5.0% / €25 / 0.5 y	
A7-ev	5.0% / 5.0% / €25 / 0.5 y	
A3a-gas	17.5% / 14.0% / €25 / 0.5 y	kwh, bill
A5b-heatpump	78.0% / 67.5% / €25 / 0.8 y	kwh, bill, payback
B2-gas_solar	5.0% / 5.0% / €25 / 0.5 y	
B2-hp_solar_batt	5.0% / 5.0% / €25 / 1.2 y	payback
B3-friend	5.0% / 5.0% / €25 / 0.5 y	
B4a-gas_solar	5.0% / 5.0% / €25 / 0.5 y	
B5-hp_solar_gridfill	5.0% / 5.0% / €25 / 1.9 y	payback
C1-gas_ev_now	19.0% / 11.5% / €60 / 0.5 y	kwh, bill, plan
D3b-gas	105.0% / 85.0% / €25 / 0.5 y	kwh, bill
D3c-gas	5.0% / 5.0% / €25 / 0.5 y	
D5e-gas	must be turned away	

How it was run

Scenarios: `python3 tests/fixtures/meter-scenarios/build.py` writes the homes, truth and files (seeded; the same bytes every time). `SCENARIO_OUT=<dir> npx playwright test meter-scenarios --grep @scenarios` runs all 71 through the app as a person would: setup, upload, the questions it asks answered from

the scenario (only from the options on screen; where none is true, the nearest), the stated system set in My system. `python3 tests/fixtures/meter-scenarios/score.py <dir>` compares with the truth. Step 0 ran the same benchmark on five builds checked out side by side, each serving its own copy of the app, with the plan list and the date frozen identically.

Limits. The homes are synthetic: their use comes from an activity model and Cork weather, their solar from that weather anchored to PVGIS. The reference battery is idealised (95% charging efficiency, no reserve), which accounts for part of the gap on the tester-like home with a file from before the panels (B1). The scenarios cover one region and one plan list. The tester's real file was not available.